

DESIGN OF THE PHYSICALLY INTERPRETABLE SAR TARGET RECOGNITION NETWORK COMBINED WITH ELECTROMAGNETIC SCATTERING CHARACTERISTICS

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ABSTRACT

Convolutional neural networks (CNNs) have been widely used for synthetic aperture radar (SAR) target recognition. However, most existing SAR target recognition methods based on CNN only use amplitude information, and the extracted feature maps are difficult to be interpreted. For SAR target, the unique electromagnetic scattering characteristics are related to its physical characteristics, such as the shape, structure and material of the target. It is of great significance to design a physically interpretable SAR target recognition network combined with electromagnetic scattering characteristics. Considering that attributed scattering centers (ASCs) can effectively characterize the electromagnetic scattering characteristics of target and component information is robust to the local variations of target, we propose to divide SAR target in the light of the geometric scattering types of ASCs and construct a multi-scale CNN to comprehensively utilizes the global information and component information of the target. Due to the utilization of ASCs, the extracted feature maps of our method are more interpretable than those extracted by traditional CNNs. Experimental results on the MSTAR dataset prove the superior performance of our method.

Index Terms— Synthetic aperture radar (SAR), target recognition, convolutional neural networks (CNN), attributed scattering center (ASC), physically interpretable.

1. INTRODUCTION

Synthetic aperture radar (SAR) can acquire high-resolution images in any weather and at any time, which has been widely used in the field of target recognition. The performance of SAR target recognition is largely affected by feature extraction. How to effectively extract features for SAR target recognition receives extensive attention.

Due to the excellent feature self-learning ability, convolutional neural network (CNN) has received widespread attention. CNN combines feature extraction and classification within a unified framework, which can

automatically learn suitable features for recognition. Many SAR target recognition methods based on CNN have been proposed. Ding et al. [1] propose to alleviate the overfitting of CNN by augmenting training samples using pose synthesis, speckle noising, and translation. Chen et al. [2] construct a novel network that only contains convolution layers, which has fewer parameters and is more suitable for SAR target recognition. Leonan et al. [3] utilize VGG-16 to effectively recognize oil rigs on SAR images.

However, most existing SAR target recognition methods based on CNN only use the amplitude information of SAR images, and the extracted feature maps are difficult to be interpreted. Actually, since the imaging mechanisms between SAR images and optical images are completely different, SAR images have unique electromagnetic scattering characteristics, which are related to the working parameters of the radar and the physical characteristics of the target, such as the shape, structure and material of the target. Therefore, it will be of great significance to design a physically interpretable SAR target recognition network combined with electromagnetic scattering characteristics.

In order to effectually characterize the electromagnetic scattering characteristics of SAR target, attributed scattering center (ASC) model [4] has been proposed. This model uses a set of parameters to characterize the frequency and azimuth dependence of the target's backscattering response. Parameters of ASC model can reflect geometric and physical attributes of target. For a SAR target, its ASCs can reflect the real physical structures of corresponding position, such as trihedral, cylinder and so on, which are more robust to the local variations of target.

For the sake of combining the ASCs and CNN to construct a physically interpretable SAR target recognition network, we propose a multi-scale CNN based on component analysis (CA-MCNN). Firstly, the SAR target is divided in the light of the geometric scattering type of ASCs, such as trihedral, cylinder and so on, to learn the component information. Then, a network that comprehensively utilize the global information extracted from entire SAR image and component information obtained from ASC model is constructed to learn the feature representation that is more robust and efficient. Finally, the obtained features will be used to classify the target. In addition, in order to enhance the description ability of features, we fuse feature maps of different scales, so that the obtained feature maps can not only contain the local and fine-grained information reflected

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by the shallower layer, but also contain the global and coarse-grained information reflected by the deeper layer.

Contributions of this paper are shown as follows:

(1) We divide SAR target in the light of the geometric scattering types of ASCs to learn the component feature maps, which makes the obtained feature maps more interpretable than those obtained by traditional CNNs.

(2) We construct a unified CNN framework that can comprehensively utilize the global information extracted from entire SAR image and component information obtained from ASC model to learn the feature representation that is more robust and efficient.

(3) We fuse feature maps of different scales to enhance the description ability of obtained feature maps.

Next, our proposed method will be introduced in Section 2. Then, Section 3 presents experimental results on the MSTAR dataset. Finally, Section 4 concludes this paper.

2. CA-MCNN FOR SAR TARGET RECOGNITION

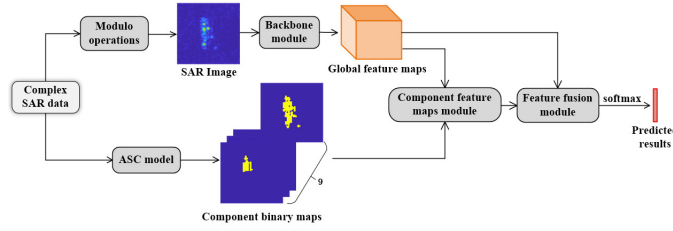


Fig. 1. The overall framework of the proposed target recognition method.

As the overall framework shown in Fig. 1, our method mainly consists of three modules, namely backbone module, component feature maps module and feature fusion module. The backbone module aims at learning global information from the input SAR image. Then, the component feature maps module will learn component information through component binary maps and global feature maps. Next, the global information and component information will be fused in feature fusion module to obtain a feature representation that is more efficient and robust, which will be used to obtain the category of the input image.

In the following sections, the three main modules of CA-MCNN will be successively introduced in detail.

2.1. Backbone module

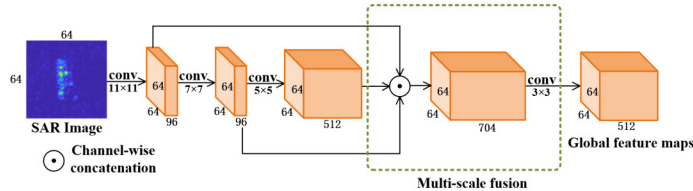


Fig. 2. The backbone module of CA-MCNN. Convolution operation is represented as “conv”. “ $m \times m$ ” represents the size of convolutional kernel.

The architecture of backbone module is shown in Fig. 2. To

ensure that the obtained global feature maps are clear enough to preserve the component information of target, we not use pooling layers in backbone module. In addition, to make input image fully learned, we use four convolutional layers in backbone module. The features of input image are extracted through the first three convolutional layers, and the last one is used to fully fuse the multi-scale feature maps to enhance the description ability of global feature maps.

2.2. Component feature maps module

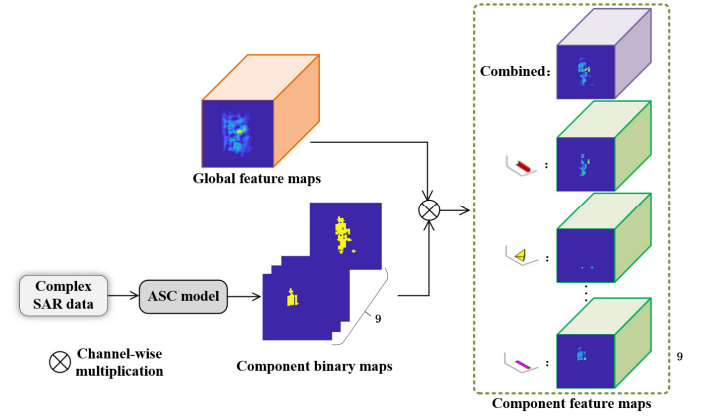


Fig. 3. The component feature maps module of CA-MCNN.

As can be seen in Fig.3, the component feature maps module has two inputs: one is global feature maps, the other is component binary maps. The component binary maps are obtained through ASC model [4]. Each ASC can be represented by a set of parameters, $\theta = [A, \alpha, x, y, L, \bar{\varphi}, \gamma]$, in which A is complex amplitude; α is frequency dependence; x and y are position coordinates of the scattering center in the range dimension and azimuth dimension, respectively; L and $\bar{\varphi}$ are length and orientation of distributed ASC, respectively; γ is aspect dependence of localized ASC.

TABLE 1. GEOMETRIC SCATTERING TYPES INFERENCE FROM FREQUENCY DEPENDENCE AND LENGTH

Geometric scattering type	Frequency dependence (α)	Length (L)	Iconic representation
Dihedral	1	> 0	
Trihedral	1	0	
Cylinder	0.5	> 0	
Top Hat	0.5	0	
Sphere	0	0	
Edge Broadside (EB)	0	> 0	
Edge Diffraction (ED)	-0.5	> 0	
Corner Diffraction (CD)	-1	0	

TABLE 2. THE DETAILED INFORMATION OF TEN-TARGET DATA

Dataset	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
Training (17°)	233 (C21)	233	232 (132)	255	299	298	299	299	299	299
Test (15°)	587 (C21, 9566, 9563)	196	582 (132, S7, 812)	195	274	274	274	273	274	274

The parameters of ASC describe the physical characteristics of target scattering centers, including relative amplitude, orientation (pose), shape and position [4]. TABLE 1 lists the correspondence between the geometric scattering types of ASC and two of these parameters. According to TABLE 1, we use the ASCs of different geometric scattering types to represent components of different types. Therefore, we can get a total of eight types of components, including trihedral, cylinder and so on.

In order to obtain the component binary maps in Fig. 3, we use the parameters of ASCs obtained by the method in [5] and the imaging parameters of the SAR data to obtain component reconstruction maps through traditional SAR imaging algorithms. Next, the component binary maps are converted from component reconstruction maps by a preset threshold. There are nine component binary maps in total, one corresponding to all components and eight corresponding to eight types of components. Finally, we multiply global feature maps with component binary maps to obtain nine sets of component feature maps, one of which corresponds to all components and eight of which correspond to eight types of components. The component feature maps describe the component information corresponding to ASC components extracted from the input SAR data, which are more interpretable than the feature maps extracted by traditional CNNs.

2.3. Feature fusion module

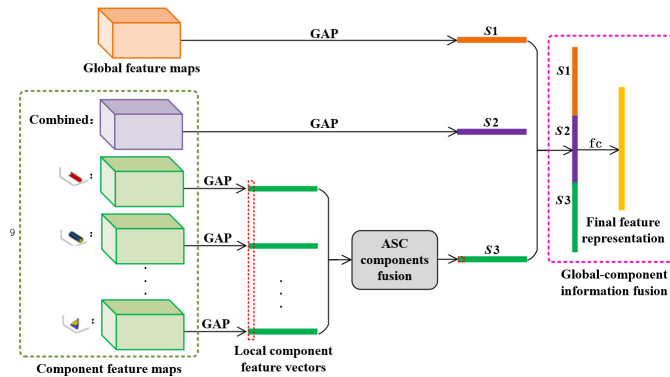


Fig. 4. The feature fusion module of CA-MCNN. “GAP” means global average pooling. “fc” indicates the fully connected operation.

As shown in Fig. 4, the feature fusion module includes ASC components fusion and global-component information fusion. Before the fusion operation, it is necessary to first convert all feature maps into feature vectors through global

average pooling (GAP) [6]. Then, ASC components fusion is performed on the corresponding positions of the eight local component feature vectors through a joint function $G(\cdot) = \max(\cdot) + \text{mean}(\cdot)$, thereby integrating them into one vector. Finally, the global-component information fusion concatenates the obtained feature vectors into one vector, and then feeds it into fully connected layer to produce the final feature representation of the target, which is more robust and effective.

3. EXPERIMENTAL RESULTS AND ANALYSIS

The MSTAR dataset with 0.3×0.3 m resolution and ten different ground military targets is used in experiments. This paper adopts two experimental data settings: ten-target data and three-target data, the details of which are listed in TABLE 2 and TABLE 3, respectively.

TABLE 3. THE DETAILED INFORMATION OF THREE-TARGET DATA

Dataset	BMP2			BTR70		T72	
	C21	9566	9563	C71	132	S7	812
Training (17°)	233	0	0	233	232	0	0
Test (15°)	196	196	195	196	196	191	195

3.1. Ablation study

In order to understand the respective effects of the fusion of multi-scale feature maps and the combination of global information and component information on performance, we design several controlled experiments in this sub-section.

TABLE 4 summarizes the experiment results on three-target MSTAR data. Compared with “Backbone module”, “Backbone module2” does not have the fusion of multi-scale feature maps. “Global information” means S1 in Fig. 5. “Component information” means S2 and S3 in Fig. 5.

TABLE 4. THE EXPERIMENTAL RESULTS OF ABLATION STUDY ON THREE-TARGET MSTAR DATA

Components?				PCC
Backbone module	Backbone module2	Global information	Component information	
×	✓	✓	×	0.9568
×	✓	×	✓	0.9407
×	✓	✓	✓	0.9788
✓	×	✓	✓	0.9861

As proved in TABLE 4, it is reasonable and effective to fuse multi-scale feature maps and combine global information with component information in CA-MCNN.

3.2. Comparisons with other target recognition methods

To demonstrate the strength of CA-MCNN, we compare it with other target recognition methods, including using the amplitude features of SAR images, Principal Component Analysis (PCA), template matching method, Dictionary Learning and Joint Dynamic Sparse Representation (DL-JDSR) [7], original auto-encoder (AE), Denoising AE (DAE), Euclidean distance restricted AE (Euclidean-AE) [8], VGGNet, and A-ConvNet [2]. All methods do not augment training data. The recognition results on three-target data and ten-target data are shown in Fig. 5 and Fig. 6, respectively. This paper adopts probability of correct classification (PCC) to measure recognition performance, which is calculated by dividing the number of correctly recognized targets by the total number of all targets.

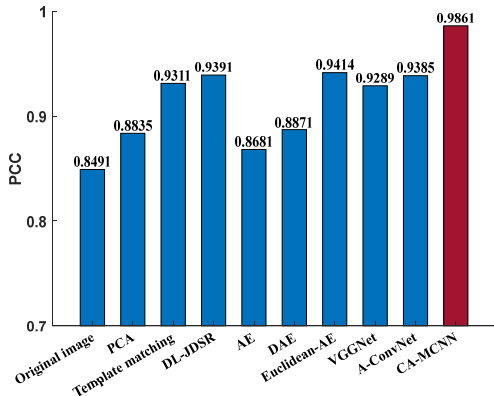


Fig. 5. Results of different target recognition methods on three-target data.

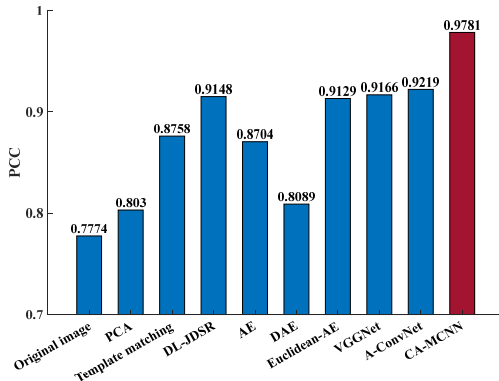


Fig. 6. Results of different target recognition methods on ten-target data.

As shown in Fig. 5 and Fig. 6, CA-MCNN has the highest PCC. As can be seen in TABLE 2 and TABLE 3, BMP2 and T72 have some variants only in test data, which are difficult to be correctly recognized by other compared methods. However, CA-MCNN comprehensively utilizes

the global information and the component information of target, the extracted features of which are more robust and efficient to the local variations of target. Therefore, CA-MCNN performs the best among all methods. The experimental results on the MSTAR dataset prove the superiority of CA-MCNN for target recognition. In addition, all experimental results also indicate that the global information and the component information of target can be better made use of by our method.

4. CONCLUSION

To overcome the defects of traditional CNN-based SAR target recognition methods, we propose a new SAR target recognition method, called the CA-MCNN method. The proposed CA-MCNN method divides the SAR target in the light of the geometric scattering types of ASCs and constructs a multi-scale CNN to comprehensively utilizes the global information and component information of the target. Since the component feature maps obtained by CA-MCNN correspond to the types of ASCs, they are more interpretable than the feature maps extracted by traditional CNNs. The experimental results based on the MSTAR dataset have shown that the proposed CA-MCNN method has the better performance for SAR target recognition than all compared target recognition methods.

5. REFERENCES

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